



A water and sediment budget for the lower Mississippi–Atchafalaya River in flood years 2008–2010: Implications for sediment discharge to the oceans and coastal restoration in Louisiana

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SUMMARY

The Mississippi is the largest riverine system in North America and one of the most engineered rivers in the world. The challenges of studying the Mississippi River are due to its complex sediment–water dynamics and the multi (and often competing) uses for its resources. Flood control and navigation are primary factors that control how the river is managed. A third factor is the use of river resources, namely water and sediment, for nourishing the degrading coastal wetlands of the states of Louisiana and Mississippi. As such, these factors must be fully considered and coordinated while investigating and developing techniques to harness the sediment resources of the River for coastal restoration. This paper presents a detailed suspended sediment budget analysis for the lowermost Mississippi and Atchafalaya River systems for the flood years of 2008, 2009, and 2010. Data were derived mainly from Federal and State of Louisiana measurements of water discharge and suspended sediment load at (1) monitoring stations along the river channel and (2) boat-based measurements made during specific project studies at natural passes and man-made channel diversions. The present study was focused on flood years 2008–2010 to (1) minimize the influence on the budgets of a historical decline in sediment loads carried by the river as observed by previous investigators and (2) take advantage of recent improvements in the monitoring network. The results show that both the Mississippi and Atchafalaya tributary pathways were efficient at sequestering suspended sediments, particularly the larger (sand) size fraction. Approximately 44% of the total Mississippi + Red River suspended load (80% of the sand) reaching Old River Control structures split between the tributaries was sequestered upstream of the Gulf of Mexico by overbank storage and channel bed aggradation. Increases in bed aggradation in the Mississippi tributary are linked to a loss of stream power associated with man-made and natural exits upstream of the Gulf of Mexico. This further decreased the water and suspended sediment load reaching the deep water Gulf tributary exits in FY08–10 to 46% (water), 19% (total suspended load), and 1.4% (suspended sand). These patterns of sediment storage and delivery have major implications for channel dredging and river sediment diversions planned for Louisiana coastal restoration.

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1. Introduction

Recent attempts to quantify the riverine sediment supply to the world ocean (Walling and Fang, 2003; Vörösmarty et al., 2004; Syvitski et al., 2005) have highlighted the degree to which river

loads have been altered by human activities in drainage basins in the past 1–2 centuries (e.g., land use changes, damming, artificial levees and other river control works, etc.). As we enter the “Anthropocene” era (Meybeck and Vörösmarty, 2005) of increasing human populations and global climate change that may alter precipitation patterns, these factors, coupled with the opposing effects on sediment discharges of many existing human alterations (for instance, dams trap sediment while deforestation increases denudation) complicate predictions of the future sediment supply of

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individual river systems. It was also recognized by Meade (1996) that quantifying the sediment supply to the world ocean by rivers can be misleading because, until recently, very few were monitored in their tidal and/or estuarine reaches due to limitations in time-series measurement instrumentation. River “end member stations” often are found 100s of km upstream of the mouth in large rivers (e.g., Amazon, 640 km; Mississippi, 492 km; Ganges, 390 km; Brahmaputra, 348 km; Meade, 1996).

The world’s 25 largest rivers drain nearly half the Earth’s land surface and account for approximately 40% of the fluvial sediments and 50% of the freshwater entering the oceans (Milliman and Meade, 1983; Meade, 1996; Vörösmarty et al., 2000). Bianchi and Allison (2009) coined the term “large-river deltaic estuary” (LDE) to refer to the portion of these large rivers that extends from the inland limit of tidal/saline penetration along lowland floodplain and deltaic plain, offshore to the limits of initial particle deposition and riverine sediment burial. Although the term was created to promote recognition that these are globally important terrigenous–marine interfaces for carbon cycling, inherent in this concept is the idea that the subaerial portion of the LDE is a location through which fluvial sediments pass prior to debouching into the ocean, and which can serve as a fluvial sediment source or trap. While trapping/sourcing likely varies in importance depending on the river due to many factors (e.g., river hydrograph, tidal range, estuarine circulation, artificial control works, etc.), our ability to examine its importance for global sediment and carbon budgets is limited by the aforementioned monitoring deficit.

Draining approximately 41% of the conterminous US, the Mississippi River has the seventh largest water discharge and suspended load of world rivers (Milliman and Meade, 1983; Meade, 1996). Sediment budgets from the upper basin to the “end member station” were first constructed (Meade and Parker, 1985; Meade et al., 1990; Meade, 1995) using monitoring data from gauging stations operated by the US Geological Survey (USGS) and the US Army Corps of Engineers (USACE). More recent budgets (Horowitz, 2010; Meade and Moody, 2010) have utilized ongoing data collections at these stations to examine declining suspended sediment loads attributed to damming and other river control works, and the multi-annual effect of major floods (such as 1993) on sediment loads. All of these budgets include monitoring station data downstream only to an “end member station”—either Tarbert Landing (492 river kilometers above the Head of Passes mouth of the Mississippi) or St. Francisville (RK 428) (Fig. 1a). Locations of the monitoring stations and other land marks along the length of the river will be identified by distance (river kilometers RK) above the Head of Passes (HOP, RK 0; Fig. 1a). The Mississippi River sediment and water budgets through the LDE reach below these stations remains poorly quantified and is the focus of the present study.

The Mississippi divides into two distributaries immediately upstream of the Tarbert Landing monitoring station at the Old River Control structures. The Old River outflow channel merges further downstream with the Red River near Simmesport, LA to form the Atchafalaya River (Fig. 1a). These two distributary channels then pass through the subaerial LDE, which extends downstream through a lowland floodplain/alluvial valley and across the Holocene-age deltaic plain (Saucier, 1994). Prior to reaching the Gulf of Mexico, the Mississippi–Atchafalaya River is modified by a network of natural and artificial water diversions, and is constrained by flood, storm, and navigation control structures, which complicate tracing water and sediment discharges through the LDE. These structures have also been cited as overarching factors in the rapid (wetland) land loss in the lower deltaic plain of coastal Louisiana (26–30 km²/y; Barras et al., 2003), both through a levee-induced reduction in overbank riverine sediment supply (Baumann et al., 1984; Walker et al., 1987; Boesch et al., 1994; Penland and

Ramsey, 1990; Swift, 1994), and by serving as pathways for tropical storm surge inundation (Barras, 2006; NAS, 2008). The continuing threat to these deltaic wetlands, compounded by accelerating sea level rise (Blum and Roberts, 2009), has led to a host of planning efforts in recent years to examine methods to stem this loss and restore coastal Louisiana (e.g., Coast2050 [<http://www.coast2050.gov>], the Louisiana Coastal Area Ecosystem Restoration [<http://lca.gov/>] plan, and post-Hurricane Katrina, the State of Louisiana [<http://www.lacpra.org>] and US Army Corps of Engineers [<http://lacpr.usace.army.mil/>] master plan documents). An understanding of the Mississippi–Atchafalaya water and sediment budget through the LDE reach is critical to the planning for engineered diversions and other mechanisms for re-introducing riverine sediments into the Louisiana coastal zone, and serves as the major motivation for the present study. The primary objective is to utilize monitoring station data and project-specific studies to examine suspended sediment fluxes and water discharge through the Mississippi–Atchafalaya LDE reach, including those portions of the fluxes passing through natural and artificial diversions prior to both distributary channels reaching the Gulf of Mexico. This budget will be utilized to examine the (1) interannual sediment storage or supply either in the channel or by overbank flow, (2) impact of tidal and estuarine processes on this budget, and (3) suitability of various reaches as sites for future river diversions.

2. Study area and background

Water movement through the Mississippi–Atchafalaya LDE reach is coordinated and funded by the USACE through the basin-wide Mississippi River and Tributaries (MR&T) Project that was initiated to minimize the impact of large floods following the catastrophic 1927 Mississippi flood. The flood control network is designed to handle a maximum Mississippi + Red River discharge of 3.03 million cubic feet per second (cfs; 85,400 cubic meters per second [cms]) at the latitude of Red River Landing, LA (RK486). This flood control involves three elements within the study area, namely (1) earthen guide levees along the Mississippi and Atchafalaya channels, (2) floodways and spillways to divert floodwaters out of the Mississippi channel above New Orleans, LA, and (3) channel improvements and bank stabilization.

Earthen levees line the entire Mississippi channel on both banks from the upper limit of the study area (Natchez, MS; RK584), except where constrained by natural bluffs. Downriver, these levees extend to RK18 at Venice, LA on the west bank, and to RK72 at Bohemia, LA on the east bank (Fig. 1b). Levee heights decrease downstream, reflecting a decreased river stage range between high and low discharge: historic stage extremes recorded by USACE at Red River Landing, LA are 0.91 and 18.1 m above National Geodetic Vertical Datum (NGVD) and at Venice are −0.2 and 2.8 m NGVD. Downriver of Baton Rouge, LA (RK372) levees are generally within 1 km of the channel, creating only narrow belts of vegetated land inside the levees (known as batture) that floods during in high discharges. Upriver of Baton Rouge, flood protection levees do not closely follow the highly meandering river channel, creating areas in tight bends where the levee set back (10–20 km from the channel) allows for significant overbank water storage during floods. However, lateral channel migration is constrained throughout the Mississippi channel in the study area through the use of concrete mats that line the channel shallows and bankline stone (rip-rap) revetments on the cut bank side of the channel. In the Atchafalaya River, the flood control levees that form the Atchafalaya Basin are set back from the channel to create a 30- to 50-km-wide flood basin. However, the upper basin (including the stations at Simmesport at RK8 and Melville at RK48) has a second set of lower earthen levees that constrain the channel to a width of less than 5 km. These

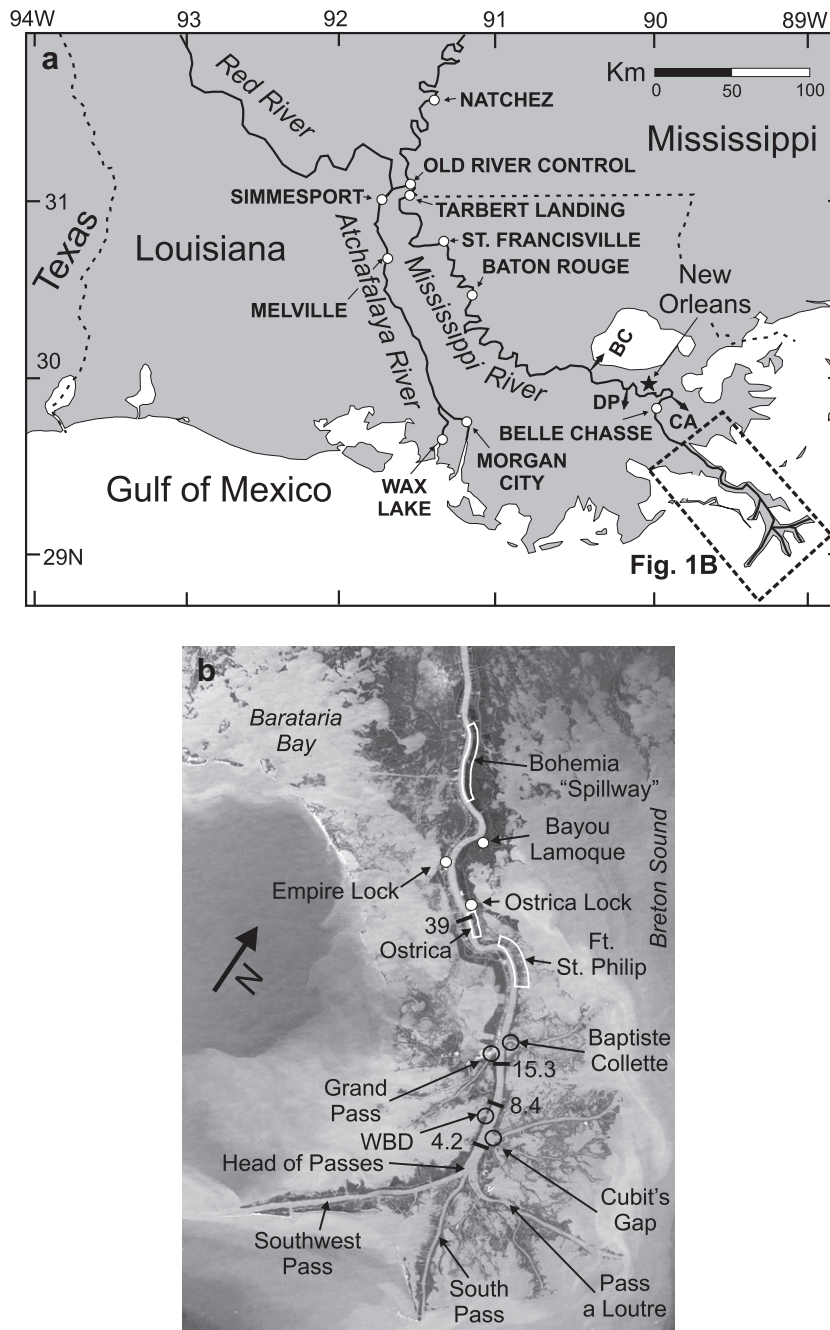


Fig. 1. (a) Map of the lower Mississippi and Atchafalaya Rivers and the location of monitoring stations discussed in the present study. Three diversion locations discussed in the text are also shown at Bonnet Carré (BC), Davis Pond (DP) and Caernarvon (CA). (b) NASA Space Shuttle visual band photograph of the lowermost Mississippi River taken on May 19, 2009 and showing the location of natural and man-made exits for water and sediment exits from the channel. Also shown are location of four river cross-sections discussed in the text at river kilometer (RK) 39, 15.3, 8.4 and 4.2.

secondary levees begin to increase in distance from the channel at about Krotz Springs, LA (RK68) and disappear below RK92 (Morgan City is at RK193). Bankline stabilization is limited to the same upper basin reach as the secondary levees, (except opposite Morgan City), where it takes the form of revetments (no mats) that are generally found on one side of the channel only, and alternate downstream between reaches sensitive to lateral channel migration.

Operation of the flood control system is coordinated by a series of discharge limits gauged at Red River Landing, LA. Regardless of water discharge, the three spillway structures at Old River Control (RK509–502) are federally mandated to remove a volume of Mississippi water, that, when added to the Red River contribution, allows

a targeted discharge of 30% of the combined discharge to pass down the Atchafalaya distributary. This contribution is adjusted daily. When discharge at Red River Landing (e.g., below Old River Control) reaches 1.25 million cfs (35,400 cms), additional water up to 250,000 cfs (7079 cms) is passed through the Bonnet Carré Spillway above New Orleans (Fig. 1a) into Lake Pontchartrain. Discharges above 1.5 million cfs (42,475 cms) cannot be passed through Bonnet Carré and are accommodated through the Morganza Spillway and West Atchafalaya Floodway upriver of Baton Rouge, and pass into the Atchafalaya Basin.

The Mississippi hydrograph exhibits a large seasonal and interannual variability, with high discharge of >28,300 cms below Old River Control occurring between Jan. and May, and typically with several

individual peaks of 1- to 2-week duration. The Atchafalaya hydrograph closely mirrors the Mississippi's given that 60–93% of the water discharge originates in the Mississippi rather than the Red River (Mossa, 1996). Mean Mississippi discharges during the high-water months are about three times the discharge during low water months (Meade, 1995). Within the LDE, the timing of the sediment discharge has been described as preceding the water discharge maxima due to hysteresis, with a more pronounced offset in large flood years (Mossa, 1996). Tides penetrate approximately 200 km into the Mississippi River during low discharge (September–November) and as little as 50 km during high discharge (January–May). Currents (0.8–1.6 m/s) in the tidal reach are dominated by the river outflow throughout the year; tidal currents are generally too weak to reverse the flow. A salt wedge is present throughout the discharge cycle, and is confined at higher discharges to the Southwest (SW) Pass channel (Fig. 1b; 14 m depth) rather than the shallower Mississippi passes to the east (e.g., South Pass, Pass a Loutre). At discharges below about 8500 cms, the saltwater wedge emerges into the main channel above the Head of Passes (HOPs; Soileau et al., 1989). The thalweg of the Mississippi River channel exceeds the depth of the surface of the Gulf of Mexico, allowing for upstream migration of the salt front for ca. 200 km above HOP, constrained and controlled primarily by the volume of water discharge, although flow duration, wind velocity and direction, tides and water temperature all are contributing factors (Soileau et al., 1989; Galler and Allison, 2008). No salt wedge penetration has been observed upriver of the main Atchafalaya entrance into Atchafalaya Bay (Galler and Allison, 2008). Significant low discharge sediment trapping of fines occurs on the channel bed in both the freshwater tidal reach of the Mississippi and associated with areas mantled by the salt wedge and driven by estuarine circulation (Demas and Curwick, 1987; Galler and Allison, 2008; Allison and Meselhe, 2010).

3. Methods

Evidence of an ongoing decline in suspended sediment load supplied from the upper basin (Meade and Moody, 2010; Horowitz, 2010), in part led to the present study being focused on creating water and sediment budgets only for flood years (FYs) 2008–2010 to minimize this systemic source of variability (although some comparison with historic data is presented as well). A flood year begins on October 1st, as this is typically close to minimum water discharge in the Mississippi River. Hence, the study period extends from October 1, 2007 to September 30, 2010.

Given the station-to-station variability in how monitoring is conducted, it was not possible to utilize a standardized methodology for calculating water and suspended sediment loads for FY08–10. Stations are divided into two types: USGS/USACE monitoring stations where a year or more of continuous data are available about water or suspended sediment discharge, and project studies, which were conducted to address a specific scientific or management issue, or which relate to a specific exit point (natural or man-made) from the river. Most of the latter results tend to extend only over a short time interval or over a few separated dates within the FY08–10 period of the budget calculations. Detailed information about the data processing for each station, including daily numbers presented as figures and as electronic files, is outlined in a [Supplementary file](#) to this paper. However, a general workflow was followed to arrive at these numbers. The first step for all stations was to calculate daily water discharge (cms). This calculation was straightforward for USGS/USACE stations making daily measurements using horizontal acoustic doppler current profiler (H-ADCP) technology (Belle Chasse, Wax Lake, Morgan City), or where the channel is well-contained within the levees leading to a high-quality gauge height:discharge water rating curve (e.g.,

Baton Rouge, for instance). H-ADCP results typically only extend across $\frac{1}{2}$ to $\frac{1}{3}$ rd to the total channel width and are extrapolated to the full cross-section (by USGS/USACE) using regular boat-based ADCP measurements that are bank-to-bank. A further reason for concentrating only on FY08–10 was that the tidally influenced daily measurement stations only have been outfitted with H-ADCP technology in recent years. At two additional monitoring stations (Melville and Natchez), stage is gauged, but discharge is not measured directly: at these stations, stage: discharge rating curves from the nearest discharge measuring stations were utilized (see [Supplementary file](#)). Water discharge at project study stations was either measured directly (by boat-based ADCP or fixed H-ADCP) or water discharge was obtained from a nearby USGS/USACE monitoring station (see [Supplementary file](#)).

Suspended sediment discharge at USGS/USACE monitoring stations typically is carried out using boat-deployed D96 or D99 (hereafter referred to as D90 series), depth-integrative, isokinetic water samplers (Edwards and Glysson, 1988) along a cross-section with 3–7 vertical profiles. The heavier sampler (D96 is 60 kg, D99 is 125 kg) is utilized at higher discharges to minimize wire angle produced by higher stream velocities in the lower Mississippi–Atchafalaya (Beverage, 1987). The USGS strategy in the lowermost Mississippi is to collect samples 12–15 times/y across the entire flood year but with denser sampling during periods of peak discharge. Depth-integrative samplers collect a single sample per vertical cast that provides a depth averaged sediment concentration (mg/l): this sample is sieved in the laboratory to determine the concentrations of individual size fractions. For the present study, the size fractions were grouped only into a sand and mud fraction (e.g., larger and smaller than 62.5 μ m fraction). Results from multiple verticals collected sequentially along the monitoring station transect are averaged to arrive at a mean cross-sectional and depth integrative concentration (Edwards and Glysson, 1988). Over the three flood-year study period, this yielded 36–45 individual data points that were then utilized to construct a suspended sediment rating curve applying the water discharge calculated for that point on that day. Separate curves were calculated for total (sand + mud) and sand load. Curve fitting software was used to test many individual best fit regression curves to these data points. Only a limited number of curve types gave the best fit (see [Supplementary file](#)) likely because of the hysteresis loop of sediment data in rivers—typically either exponential rise to maximum, power law, or logarithmic curves.

Two monitoring stations (Baton Rouge and Belle Chasse) and several project study sites used optical backscatterance sensors (OBSs) to obtain daily-averaged turbidity (NTU) measurements. These turbidities were calibrated to total (mud + sand) suspended sediment concentration using a ratings curve developed from the isokinetic boat data for available days (see [Supplementary file](#)) in FY08–10. Other project study sites were sampled using point-integrative isokinetic suspended sediment samplers (P-61 at 46 kg or P-63 at 91 kg). Where multiple depths were measured in a vertical, these values were averaged to be comparable with depth-integrative sampler results from the monitoring stations. The cross-sectional averaged concentrations (mg/l) were then fit to a sediment rating curve with boat-measured ADCP water discharges in the same way as the monitoring station data (see [Supplementary file](#)). Sample comparisons between the heavier P-63 with D90 results agreed well since both were deployed to a maximum of 0.9 total water depth; when the P-61 was utilized (e.g., Natchez, MS), suspended loads were significantly lower—likely because of an increase in wire angle which reduced true depth (Beverage, 1987). Where suspended sediment measurements were not available at project study sites, local water discharges were combined with concentrations for that day at the nearest USGS/USACE monitoring station to estimate loads.

For the Mississippi reach below the lowest monitoring station (Belle Chasse), channel sediment storage was calculated by a different method. The two most recent USACE decadal navigation surveys from January–March 1992 and August 2003 were utilized to calculate shoaling over specific bank-to-bank reaches extending from RK121 (Belle Chasse) to RK4.2. Volumes were corrected to sediment loads (MT/y) using methods outlined in the [Supplementary file](#).

4. Results

4.1. Channel monitoring stations

Annual water discharge during the three flood years (FY08–10) ranged from 479 to 607 cubic km (Table 1) at the Tarbert Landing station below Old River Control Structures (ORCSs), with maximum daily discharges of 41,229 (FY08), 36,161 (FY09), and 28,883 (FY10) cms. FY10 had an unusual hydrograph of five individual freshets (see [Supplementary file](#) figures for individual station water discharge records), including record high levels in October 2010, which led to the largest annual flow volume for the 3 years despite the smallest maximum discharge. Maximum discharges in FY08 took place during an opening (April 12–May 7, 2008) of the Bonnet Carré Spillway above New Orleans triggered by a discharge at the Red River Landing stage gauge near Tarbert Landing of 1.25 million cfs (35,396 cms). The 2008 event was the 9th occasion that the Bonnet Carré structure had been opened since 1930. Annual water discharge at Tarbert Landing in 1950–2007 ranged from 220 to 665 cubic km (Meade and Moody, 2010), and all 3 years of the present study were above the median: 4th, 8th and 24th highest of the 60-year time span (adding these years after the published study).

Averaged over the three flood years (Fig. 2), water discharge trends downriver exhibited a significant variability—particularly in the Mississippi channel. Water discharge at Natchez, when corrected for the water loss through ORCS (167 cubic km/y) was 15 cubic km/y lower than Tarbert Landing, implying an additional source(s) of water in the Natchez to Tarbert Landing reach. At stations below Tarbert Landing on the Mississippi this trend was reversed, with discharge decreasing by 25 cubic km/y between Tarbert Landing and Belle Chasse. A calculated seven cubic km/y of this loss was through Bonnet Carré (open and closed), and through the freshwater diversion structures at Davis Pond and Caernarvon (Fig. 1). No water discharge data are presented for the St. Francisville station because overbank flow complicates rating the local stage measurement (see [Supplementary file](#); sediment data are calculated based on Baton Rouge water). In the Atchafalaya channel, 70 cubic km/y of Red River water (Fig. 2) are needed

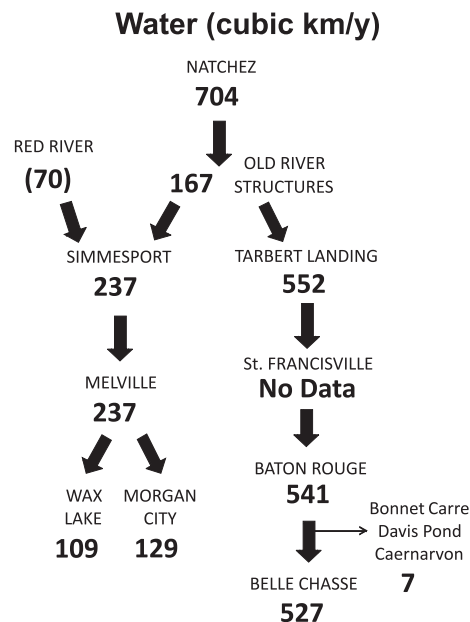


Fig. 2. Average annual water discharge (in cubic km/y) for the three flood years (2008–2010) discussed in the present study for US Geological Survey and US Army Corps of Engineers monitoring stations in the lower Mississippi and Atchafalaya Rivers. Input from the Red River is calculated indirectly (Simmesport water minus water discharge through the Old River Structures). Integrated water loss for the three diversion channel exits between Baton Rouge and Belle Chasse are also shown.

to account for the measured 3-year average water discharge at Simmesport, given the Mississippi contribution through ORCS. This number is a 29.5% Red + Mississippi water contribution, very close to the federally mandated 30% Red + Mississippi discharge. Below Simmesport, water discharge was unchanged at Melville, and is relatively unchanged (+1 cubic km/y) in the combined Wax Lake and Morgan City outflow (238 cubic km/y; Fig. 2).

Fig. 3 displays a downriver budgeting of total and sand (>62.5 μ m) suspended sediment recorded at the USGS/USACE monitoring sites averaged over the FY08–10 period. Table 1 lists the annual values used to calculate these 3 year site averages. The values in Fig. 3 are not presented with error bars (representing variability between the individual years) as that would inflate the apparent measurement error. As Table 1 shows, year-to-year increases or decreases in suspended sediment load at a station are mimicked at all the stations. Total suspended discharge at Tarbert Landing in FY08–10 ranged from 137 to 178 million metric tons (MT): Meade and Moody (2010) reported a 20-year averaged

Table 1

Water and sediment discharge in FY08–10 for USGS/USACE monitoring stations in the lower Mississippi and Atchafalaya Rivers.

Station	Annual water discharge in cubic km/y (in 10^{11} ft ³ /y)			Total sediment discharge in 10^6 metric tons/y (in 10^6 short tons/y)			Sand discharge in 10^6 metric tons/y (in 10^6 short tons/y)		
	FY2008	FY2009	FY2010	FY2008	FY2009	FY2010	FY2008	FY2009	FY2010
Natchez	712 (252)	608 (215)	792 (280)	Undefined	Undefined	Undefined	Undefined	Undefined	Undefined
Tarbert Landing	570 (201)	479 (169)	607 (215)	155 (171)	137 (152)	178 (197)	73.5 (81.0)	62.2 (68.6)	78.9 (86.9)
St. Francisville	No data	No data	No data	91.0 (100)	75.8 (83.5)	104 (115)	17.6 (19.4)	14.4 (15.8)	21.0 (23.2)
Baton Rouge	548 (193)	471 (166)	605 (214)	93.3 (103)	81.0 (89.3)	103 (114)	31.8 (35.1)	20.1 (22.2)	31.6 (34.8)
Belle Chasse	523 (185)	465 (164)	592 (209)	88.7 (97.8)	72.7 (80.1)	103 (114)	18.6 (20.5)	12.7 (14.0)	22.1 (24.4)
Simmesport	244 (86.2)	206 (72.7)	260 (92.0)	72.8 (80.3)	61.7 (68.0)	80.0 (88.2)	16.8 (18.5)	10.8 (11.9)	15.9 (17.5)
Melville	243 (86.0)	206 (72.8)	261 (92.1)	39.0 (43.0)	30.8 (34.0)	42.9 (47.3)	13.6 (15.0)	8.4 (9.3)	12.7 (14.0)
Morgan City	133 (47.1)	113 (40.0)	140 (49.4)	30.9 (34.1)	22.7 (25.1)	30.1 (33.2)	7.3 (8.0)	3.8 (4.2)	5.0 (5.5)
Wax Lake	108 (38.0)	95 (33.5)	126 (44.4)	20.1 (22.1)	16.9 (18.6)	24.6 (27.1)	2.3 (2.6)	1.4 (1.5)	2.5 (2.7)
RK39	520 (184)	463 (164)	588 (208)	88.0 (97.1)	73.8 (81.4)	97.0 (107)	No data	No data	No data
RK8.4/15.3	133 (119)	304 (107)	378 (133)	31.2 (34.4)	23.9 (26.3)	36.8 (40.6)	4.1 (4.5)	2.3 (2.5)	2.9 (3.2)
RK4.2	275 (97.1)	244 (86.1)	312 (110)	31.8 (35.0)	24.7 (27.2)	37.6 (41.5)	2.3 (2.5)	1.2 (1.3)	1.4 (1.5)

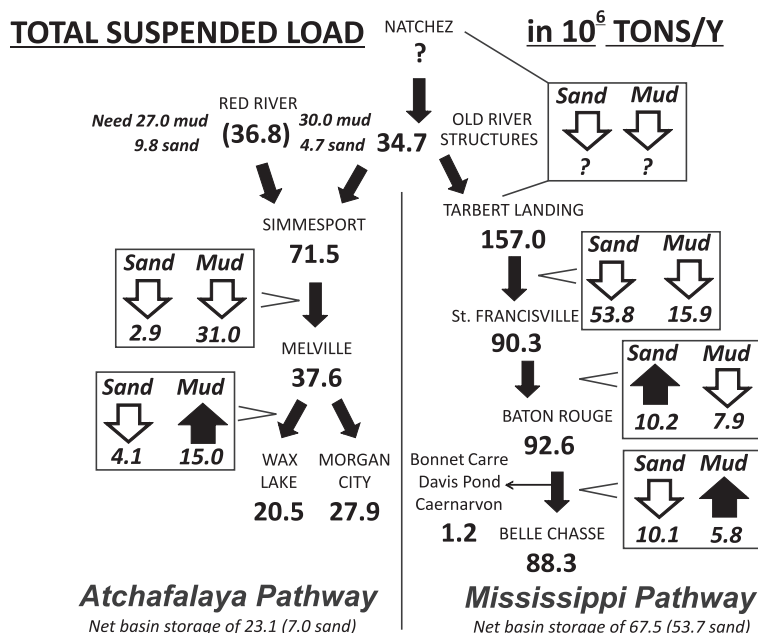


Fig. 3. Average annual total sediment discharge (in 10^6 tons/y) for the three flood years (2008–2010) discussed in the present study for US Geological Survey and US Army Corps of Engineers monitoring stations in the lower Mississippi and Atchafalaya Rivers. Input from the Red River is calculated indirectly (Simmesport minus sediment discharge through the Old River Structures). Integrated sediment loss for the three diversion channel exits between Baton Rouge and Belle Chasse are also shown. Arrows within boxes (white down = loss from suspended load, black up = addition to sediment load) indicate changes in suspended sediment load (in 10^6 tons/y) between stations separated into sand ($>62.5 \mu\text{m}$) and mud ($<62.5 \mu\text{m}$) fractions. Net basin sediment storages for the lowermost Mississippi (load differential between Tarbert Landing and Belle Chasse) and Atchafalaya (load differential between Simmesport and Morgan City + Wax Lake) are also shown.

(1987–2006) annual total suspended load at Tarbert Landing of 115 MT. Atchafalaya total sediment discharge at Simmesport was 62–80 MT/y in FY08–10, compared to a 1987–2006 average (Meade and Moody, 2010) of 57 MT/y. St. Francisville in FY08–10 ranged from 76 to 104 MT/y, compared to a 1993–2007 average of 84 MT/y calculated by Horowitz (2010). Data at all three stations suggest flood years 2008 and 2010 were above average (following the trends for water discharge noted above) for total suspended sediment discharge and 2009 was near, to slightly above average.

The historical sediment discharge records at Tarbert Landing, St. Francisville, and Simmesport examined by Horowitz (2010) and Meade and Moody (2010) indicate progressively declining sediment loads since the 1950s. They attribute that decline to the integrative effects of catchment dams, soil erosion control, river training and bank revetment structures, and a supply limitation caused by large floods such as 1993. Fig. 4 are plots of isokinetic, boat-collected discharge data from 1978 to 2010 relative to the ratings curve data utilized to estimate sediment loads (total and sand) in FY08–10 at the two stations (Belle Chasse and Morgan City) furthest downstream in the Mississippi and Atchafalaya. Total sediment loads relative to water discharge were significantly higher in 1978–1987 at both stations. The decrease in sand loads displayed by FY08–10 data were confined to lower discharges ($<15,000$ cms in the Mississippi and 4000 cms in the Atchafalaya) and were more pronounced at Belle Chasse. These data suggest that the declining sediment loads observed by previous investigators propagate into the lowermost Mississippi–Atchafalaya as well.

The suspended sediment budget (total and sand) for FY08–10 in Fig. 3 displays large downstream fluctuations in sediment load in both the Mississippi and Atchafalaya. Data for Natchez, Mississippi are not shown because of sampler bias (see Supplementary file) associated with the use of a lighter sampler and a maximum sampling depth of 0.84 total water depth (other stations were 0.9). There was a large decrease in sand load (54 MT) in the Mississippi between Tarbert Landing and St. Francisville. Sand discharge increased by

10 MT/y at Baton Rouge and declined by a similar amount at Belle Chasse. Mud ($<62.5 \mu\text{m}$) discharge declined from Tarbert Landing to Baton Rouge and increased slightly (6 MT/y) at Belle Chasse. Both sand and mud load decreased in the Atchafalaya between Simmesport and Melville, with mud loss three times that of sand. Sand loads continued to decrease in the combined Morgan City and Wax Lake outlet stations to Atchafalaya Bay, but mud load increased by 15 MT/y. The lowermost river pathways, taken as Tarbert Landing to Belle Chasse (Mississippi) and Simmesport to Morgan City + Wax Lake, displayed a net basin storage (Fig. 3) of 68 and 23 MT/y, respectively. Of that total, 80% was sand in the Mississippi reach and 30% in the Atchafalaya reach. This loss is that fraction recorded at Tarbert Landing and Simmesport that does not reach below the lowermost monitoring stations (net annual basin storage). In the Atchafalaya, the lowermost stations are immediately adjacent to the coast (Atchafalaya Bay), while Belle Chasse is located at RK121 above Head of Passes (HOP; RK0).

An operational definition used by the USGS and USACE in Louisiana of when the lower Mississippi River is in “flood” (high discharge) is 700,000 cfs (19,822 cms), while “low” flows are those below about 535,000 cfs (15,150 cms). To examine the effects of sediment throughput between stations with the water discharge hydrograph, Fig. 5 is a plot of the annual suspended sediment load (mud and sand) carried at water discharges above and below these values. The discharge interval between “low” and “high” is considered a “transitional” (rising or falling phase) hydrograph. Atchafalaya low, transitional, and high values at Simmesport and Melville are reduced to account for the FY08–10 water fraction carried by Simmesport versus Tarbert Landing (42.9% annual water discharge; Fig. 2) so “low” is $<230,000$ cfs and “high” is $>300,500$ cfs. Values are reduced proportionally again to account for the flow division between Morgan City (54.2% of water) and Wax Lake (45.8%). The results show that the suspended sediment load (sand and mud corrected to tons/d), increased from low to high discharge at all eight stations. In the four Mississippi stations,

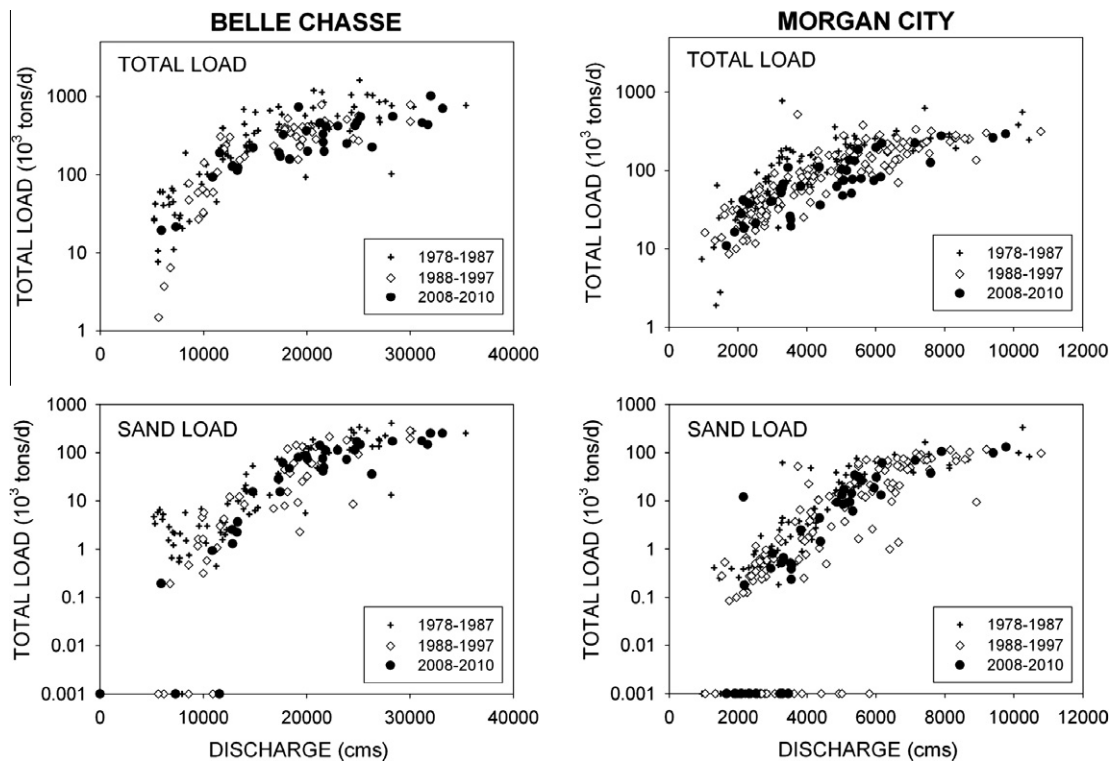


Fig. 4. Depth-integrated isokinetic sampler, boat-based measurements of suspended sediment load (total in upper plots, sand load in lower) collected by the US Geological Survey at two lowermost stations in the Mississippi (Belle Chasse) and Atchafalaya (Morgan City) rivers. The 2008–2010 data points are those utilized to calculate the suspended loads reported in the present study. These are compared with data collected at the same site in previous decades (1978–1987 and 1988–1997) using the same methodology.

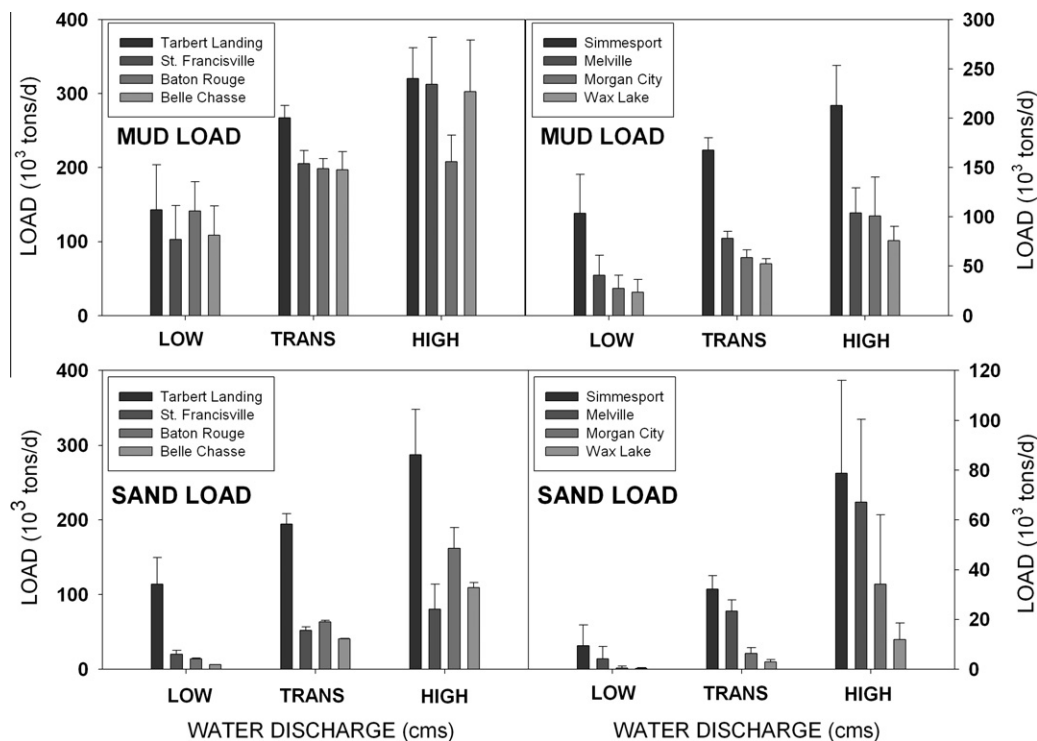


Fig. 5. Mud (<62.5 μm ; upper plots) and sand (lower plots) suspended sediment loads at four Mississippi (left plots) and four Atchafalaya (right plots) monitoring stations in flood years 2008–2010 for low, transitional and high water discharge phases of the hydrograph. Figures are calculated in tons/d utilizing mean (and standard deviation error bars) values for all days over the 3 years that were within the flow range. Flow divisions are discussed in the text.

daily mud loads increased by a factor of 1.5 (Baton Rouge) to 3.0 (St. Francisville) between low and high discharge. Sand loads were greater at high discharge by a factor of 2.5 at Tarbert Landing, increasing progressively downstream to 18.5 at Belle Chasse. In the Atchafalaya stations, daily mud loads increased by a factor of 2.1 (Simmesport) to 3.7 (Morgan City) between low and high discharge: sand loads were greater at high discharge by a factor of 8.4 at Simmesport, increasing progressively downstream to 53.6 at Wax Lake and 81.4 at Morgan City.

4.2. Water and sediment loss in the Belle Chasse to Head of Passes Mississippi River reach

Three-year means of annual water losses from the Mississippi River channel below the Baton Rouge and Belle Chasse monitoring stations are quantified in Fig. 6 and Table 2. These water discharges are quantified from individual ratings curves for each channel exit, developed from measurements taken during project studies in FY08–10 (see [Supplementary file](#)). Several other smaller types of water losses are not included in this assessment. Water siphons utilized for small freshwater diversions are present at Naomi (RK103) and West Point a la Hache (RK79). They are not included because (1) of their small maximum discharge (56.6 cms), (2) they only operate at higher discharges, and (3) operational record keeping is non-continuous. Navigational locks at Ostrica (RK40), Empire (RK47), Intracoastal Waterway-Algiers (RK142), Intracoastal Waterway-Harvey (RK158) and New Orleans Industrial Canal (RK150) also are not included because of their (1) small maximum discharge, and (2) the difficulty in quantifying the number of opening cycles and volumes of water exchanged in each cycle. An older freshwater diversion is also present at Bayou Lamoque (RK54) that consists of two structures with four box culverts each, and a rated discharge of 28.3 cms for each culvert. However, this structure is operated locally, with limited record keeping and measurements conducted by the authors (Allison and Vosburg) during the Gulf

Horizon Oil Spill in 2010 when the structure was opened by the State to full capacity showed discharges less than half the rated maximum.

Approximately 50% of the average annual water loss between Baton Rouge and Belle Chasse (14 cubic km/y) can be accounted for by the Bonnet Carré (open in 2008 only), Davis Pond and Caernarvon diversions. Below Belle Chasse, water loss through *measured* natural and artificial exits above HOP totaled 236 cubic km/y, or 44.8% of flow. An independent measurement of *total* water loss in the reach can be arrived at by summing the total flows of the three river mouth passes (e.g., Southwest, South, Pass a Loutre) which were measured individually and a ratings curve constructed relative to Belle Chasse (see [Supplementary file](#)). These measurements were used for the values in Fig. 6 and total 253 cubic km/y, or 48% of the volume at Belle Chasse. These calculations also indicate Southwest Pass carried 64% of the total water discharge below HOP.

Suspended sediment loss (total and sand) is shown in Fig. 7. Only 4.6% of the average annual suspended sand loss between Baton Rouge and Belle Chasse (10.1 MT/y) can be accounted for by the Bonnet Carré (open in 2008 only), Davis Pond, and Caernarvon diversions. Given that the mud load increased between Baton Rouge and Belle Chasse, the three diversions discharged only 0.2 MT/y out of the total increase of 6.0 MT/y of mud load between stations. Below Belle Chasse, sediment loss through measured natural and artificial exits above HOP totaled 29.4 MT/y (33%), including 4.6 MT/y (26%) of sand (see [Supplementary file](#) for methods of calculating sand loss through each exit).

4.3. Channel storage in the Belle Chasse to Head of Passes Mississippi River reach

Suspended sediment load loss or gain calculated for specific river reaches (Fig. 3) does not encompass the lowermost Mississippi region from Belle Chasse (RK121) to HOP. For this reach, channel sediment storage was calculated by a different method; utilizing USACE navigational surveys from 1992 and 2003 below the water line, and deriving MT/y from shoaling displayed in specific river reaches. No more recent mapping was available to do a comparison during the FY08–10 period of examination for other stations. Hence, any intercomparison of these data with the monitoring station results presented in the following sections has the caveats that the channel storage quantities reflect an accretion that is (1) over an earlier period than the station data and (2) encompasses an entire decade (including a 1997 Bonnet Carré opening). The channel storage was divided into eight reaches (density assumptions and results shown in [Supplementary file](#) and plotted in Fig. 8), and are combined into three individual reaches in Fig. 6. The storage in Fig. 7 was 8.5 MT/y integrated over the entire Belle Chasse to RK4.2 reach. Fig. 8 demonstrates a downriver trend of channel bed storage of sediment threefold increasing when plotted in terms of tons per river kilometer (RK).

5. Discussion

5.1. Lower Mississippi–Atchafalaya basin storage in the monitored reach

The suspended sediment load calculations for the monitored reach of the lower Mississippi (Tarbert Landing to Belle Chasse) and Atchafalaya (Simmesport to Morgan City/Wax Lake) were potentially impacted by variability in annual water discharge (Fig. 2, Table 1). Such variability in the water discharge influences the sediment ratings curves. In the Atchafalaya, calculated annual water discharge averaged over FY08–10 varied by only 1 cubic km

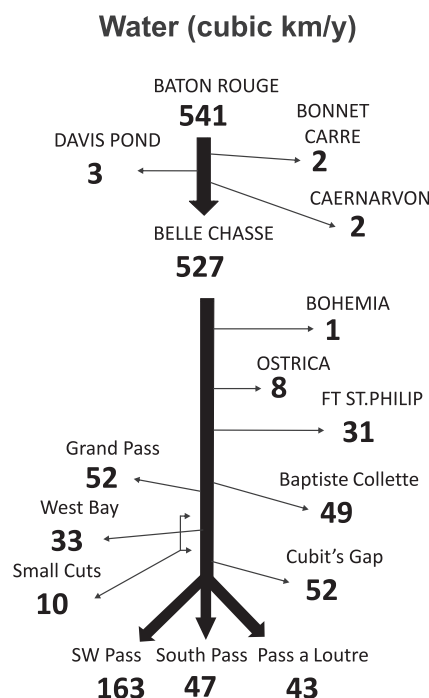


Fig. 6. Average annual water discharge (in cubic km/y) for the three flood years (2008–2010) discussed in the present study for natural and man-made water exits from the Mississippi River below Baton Rouge, Louisiana. Locations of these exits are shown in Fig. 1.

Table 2

Water and sediment discharge in FY08–10 for channel diversions and passes in the lower Mississippi River.

Station	Annual water discharge in cubic km/y (in 10^{10} ft ³ /y)			Total sediment discharge in 10^5 metric tons/y (in 10^5 short tons/y)			Sand discharge in 10^5 metric tons/y (in 10^5 short tons/y)		
	FY2008	FY2009	FY2010	FY2008	FY2009	FY2010	FY2008	FY2009	FY2010
Old River Control	187 (662)	141 (498)	172 (607)	403 (444)	285 (314)	352 (388)	64.3 (70.9)	32.8 (36.1)	43.4 (47.9)
Bonnet Carré	5.7 (20.3)	0.2 (0.7)	0	9.4 (10.4)	0.4 (0.4)	0	0.6 (0.7)	0.2 (0.2)	0
Davis Pond	2.5 (8.9)	2.9 (10.4)	3.5 (12.3)	4.4 (4.8)	4.6 (5.0)	5.7 (6.2)	0.9 (1.1)	0.8 (0.9)	1.0 (1.0)
Caernarvon	2.3 (8.0)	1.4 (4.8)	2.7 (9.4)	4.0 (4.4)	1.9 (2.1)	4.4 (4.8)	0.9 (1.0)	0.3 (0.3)	0.8 (0.9)
Bohemia	1.7 (5.9)	0.9 (3.3)	0.9 (3.3)	3.7 (4.1)	2.0 (2.3)	2.0 (2.2)	1.1 (1.3)	0.6 (0.7)	0.6 (0.7)
Ostrica	9.9 (35.0)	4.5 (16.0)	10.7 (37.8)	21.0 (23.2)	9.8 (10.8)	21.8 (24.1)	6.2 (6.8)	2.9 (3.2)	6.0 (6.7)
Ft. St. Philip	29.9 (106)	24.7 (87.2)	37.8 (134)	56.3 (62.0)	42.9 (47.3)	68.5 (75.6)	13.5 (14.9)	8.9 (9.8)	15.4 (17.0)
Baptiste Collette	48.9 (173)	42.9 (152)	56.0 (198)	88.4 (97.4)	76.0 (83.8)	103 (114)	18.6 (20.5)	10.3 (11.3)	15.6 (17.2)
Grand Pass	51.7 (183)	46.4 (164)	58.0 (205)	51.6 (56.8)	39.3 (43.3)	60.2 (66.3)	7.9 (8.7)	4.6 (5.1)	7.1 (7.9)
West Bay	32.3 (120)	28.5 (95.1)	36.8 (130)	32.7 (36.0)	24.1 (26.6)	39.0 (43.0)	3.8 (4.2)	2.1 (2.3)	2.6 (2.9)
Small Cuts	10.4 (36.9)	10.3 (36.2)	10.6 (37.4)	10.1 (11.2)	8.7 (9.6)	11.9 (13.1)	1.1 (1.3)	0.6 (0.7)	1.1 (1.3)
Cubit's Gap	51.2 (181)	43.6 (154)	60.3 (213)	38.6 (42.5)	26.3 (29.3)	44.5 (49.0)	1.6 (1.8)	0.9 (1.0)	1.1 (1.2)
Southwest Pass	161 (570)	137 (484)	190 (671)	212 (234)	160 (177)	251 (276)	16.5 (18.2)	8.9 (9.8)	10.1 (11.1)
South Pass	47.0 (166)	42.0 (148)	52.9 (187)	46.9 (51.7)	37.9 (41.8)	55.0 (60.7)	3.2 (3.5)	1.7 (1.9)	2.0 (2.2)
Pass a Loutre	42.5 (150)	39.3 (139)	46.2 (163)	47.8 (52.7)	39.1 (43.1)	55.9 (61.7)	3.1 (3.4)	1.7 (1.9)	2.0 (2.2)

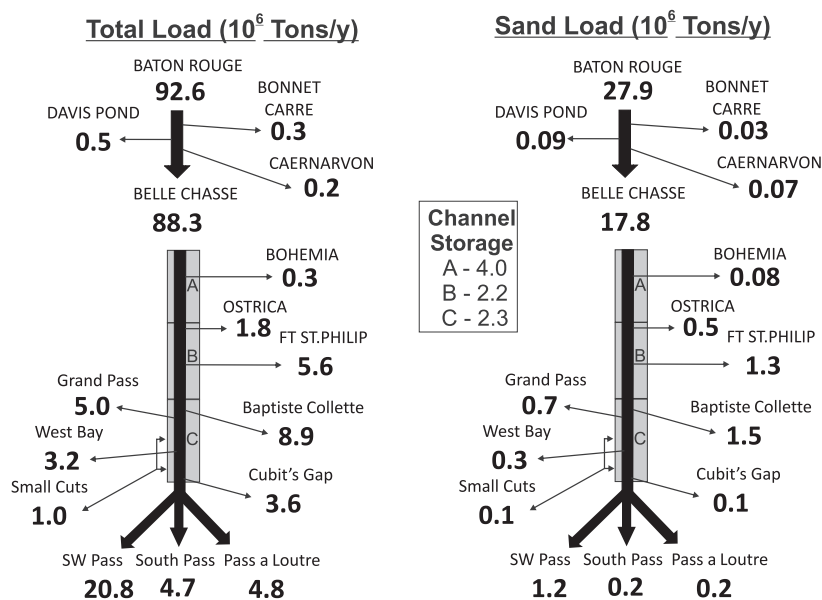


Fig. 7. Average annual suspended sediment discharge (total suspended load at left, sand load at right (in 10^6 tons/y) for the three flood years (2008–2010) discussed in the present study for natural and man-made water exits from the Mississippi River below Baton Rouge, Louisiana. Also shown are annual channel storage rates (in 10^5 tons/y) for three sub-reaches (A–C) of the channel between Belle Chasse and river kilometer (RK) 4. Rates were calculated bathymetric changes derived from decadal navigational surveys as described in the text.

(<0.5%) over the study reach. In the Mississippi, however, there was a progressive water loss between Tarbert Landing and Belle Chasse (corrected for diversion loss) of 18 cubic km/y. An undetermined amount of loss likely occurred due to withdrawal of drinking water by cities (New Orleans and smaller communities downstream of Baton Rouge, which utilizes groundwater from wells). This loss may be balanced in part by wastewater discharge returned to the river. Losses above this amount would be through industrial uses, as well as evaporative flux and discharge to groundwater. While this water loss likely had some impact on sediment transport capacity, the water loss is smaller than the individual loss from five of the channel exits below Belle Chasse (Fig. 6).

In the Tarbert Landing to Belle Chasse reach of the lower Mississippi, there is a net annual loss from suspended sediment load of 67.5 MT/y, or 43% of the total (Fig. 3). Two possibilities exist to explain this basin storage: channel aggradation or loss to overbank flow. A study of channel aggradation and deepening of the Mississippi River reach between Arkansas City, Arkansas (RK917) and

ORCS for the period 1949–1989 is summarized in Biedenharn et al. (2000) and Harmar et al. (2005). These studies utilized water surface slope (from stage gauges) and decadal USACE channel bathymetry surveys to discover that a major increase in stream power took place post-1949 associated with channel straightening, bank stabilization, and channel narrowing produced by lateral channel dyke-field construction. Water surface slope increased throughout the reach (Biedenharn et al., 2000) and Harmar et al. (2005) conclude that the reach upstream of Vicksburg, MS (RK702) responded by rapid channel deepening (downcutting). This deepening trend became negligible after about 1975, presumably because the Mississippi had achieved a post-cutoff equilibrium. From Vicksburg to the ORCS, the longitudinal bed profile downcut slightly pre-1975 but in 1975–1989 began aggrading (Biedenharn et al., 2000). As these results measure aggradation only extending to the ORCS, they do not encompass our study reach immediately below the structures—direct comparison is hindered by the sampler bias in the results from Natchez, MS station

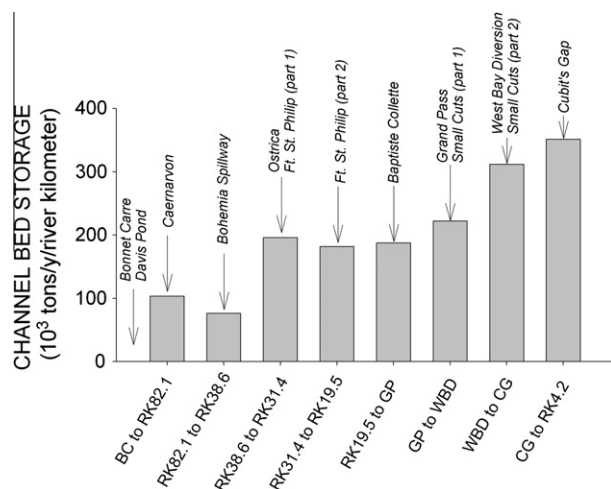


Fig. 8. Annual channel storage rates (in 10^6 tons/y) divided into eight sub-reaches of the Mississippi River channel between Belle Chasse and river kilometer (RK) 4. Rates were calculated bathymetric changes derived from decadal navigational surveys as described in the text. Also plotted are the sub-reach locations of major water and sediment exits from the river included in the present study.

(see [Supplementary file](#)). Another factor which favors channel aggradation in this reach is the “backwater effect” ([Parker, 2004](#); [Chaudhry, 2008](#)), whereby stream power is reduced as a river approaches its receiving basin and transitions from gravity (slope)-driven to mixed gravity and pressure-driven transport. [Nitttrouer et al. \(2011a\)](#) demonstrate in the lower Mississippi this backwater transition is found at lower discharges in the reach from HOP to about RK650—a divergence between water surface and bed slopes begins immediately above Natchez (RK584). The large storage loss observed in the present study between Tarbert Landing and St. Francisville coincides with the expected transition zone, and the predominance of sand storage suggests a hypothesis that fines stored at lower discharges were preferentially remobilized by the gravity-driven high flow events ([Horowitz, 2010](#)).

A detailed stage and bathymetric survey examination has not been conducted to date for the ORCS to Baton Rouge reach, or for post-1989. However, [Galler et al. \(2003\)](#) analyzed the Baton Rouge to HOP reach decadal bathymetric surveys from 1893 to 1992 and found net channel aggradation from Baton Rouge to RK175, with a switch to deepening to HOP. The change was mainly in the period 1921–1948 and was ascribed to the channel adjustment associated with the construction of federal guide levees in 1928–1930. They report the reach has been relatively stable post-1948 in contrast to the present evidence of aggradation below Belle Chasse ([Fig. 8](#)).

A final factor favoring channel aggradation in the Tarbert to St. Francisville reach is the loss in stream power below the ORCS caused by the exit of water into the Atchafalaya Basin. The stream power reduction can be observed in the rapid drop in river surface slope passing the ORCS. Utilizing an average stage over FY08–10 for each station, the reach from Natchez to Knox Landing RK585) immediately above the ORCS has an average slope of 0.118 m/km, from Knox Landing to Red River Landing; below the ORCS (RK487) the slope is 0.125 m/km, while that from Red River Landing to Baton Rouge is 0.101. It should be noted that the overall trend of sand loss during transport is interrupted in the St. Francisville to Baton Rouge reach ([Fig. 6](#)). A possible explanation for this is anthropogenic: maintenance dredging in the channel at Baton Rouge Harbor (RK378.5) and Baton Rouge Front (RK375). A total of 1.38 million m^3/y were dredged from the channel margin in FY08–10 and dumped by hopper dredge into the channel thalweg immediately upriver and adjacent to the Baton Rouge monitoring station (RK370.1) (E. Creef, pers. comm.).

Overbank flooding into the vegetated batture region inside the guide levees cannot be ruled out as an additional source of loss from suspended load in the lower Mississippi reach. Evidence of this is the difficulty of calculating daily water discharge at St. Francisville from the nearby stage gauge at Bayou Sara (outlined in [Supplementary file](#)), which stems from the overbank flow at this site at discharges above 21,230 cms. Although the extent of this overbank batture region is still constricted relative to the pre-levee floodplain (average width 4.8 km; [Biedenharn and Thorne, 1994](#)), the region above Baton Rouge is of much greater extent than further downriver, where guide levees are within 1 km of the river. This mechanism also may explain the storage of mud observed from Tarbert Landing to Baton Rouge ([Fig. 6](#)).

In the Simmesport to Morgan City/Wax Lake reach of the lower Atchafalaya, there was also a net annual loss from suspended load of 23.1 MT/y or 32% of the total ([Fig. 3](#)). Unlike the lower Mississippi, the majority of the loss was mud (30% sand). The reduction in stream power approaching the Gulf of Mexico associated with the backwater effect, outlined for the lower Mississippi, also may play a role in the Atchafalaya as well. However, the 97% duration stage: discharge relationship at Morgan City was calculated for the four low discharges in FY08–10: a 1642 cms discharge corresponded to a stage of 0.77 ± 0.31 m. An analysis of stages at Simmesport and Melville during the 97% duration stage in 1980–2010 (not shown), exhibited an interannual variability in stage height that can be attributed to seasonal hysteresis ([Simons et al., 1973](#)) since this duration occurred in July–December in various years. More significantly, these interannual trends were mirrored at Simmesport and Melville, suggesting limited channel deepening or aggradation at either site has altered the water surface slope at a given discharge relative to Morgan City through time.

A more likely explanation is overbank storage. The Atchafalaya Basin width ranges from 30 to 50 km in the lower basin (Krotz Springs to Morgan City/Wax Lake) to less than 5 km in the upper basin, allowing for more significant overbank storage. It is unclear whether the secondary levees in the upper basin constrain overbank flooding at the highest discharges, or whether water exits through small, controlled and uncontrolled canals and bayous into the wider basin floodplain. The overbank storage mechanism would tend to support a less sand-dominated removal from suspended transport. [Hupp et al. \(2008\)](#) utilized feldspar marker horizons to document 2–42 mm/y sediment accumulation rates in the Atchafalaya Basin floodplain in an area between Melville and Morgan City in 2000–2003, with 5–44% of the storage composed of sand. Lacustrine deltas progradation into the lake reaches of the basin is also well documented as a process that stores Atchafalaya sediment ([Tye and Coleman, 1989](#)), particularly sand. Although many of these lakes are now infilled, which has been cited as a reason for bypassing to Atchafalaya Bay and subaerial bay head delta emergence after 1973 ([Rouse et al., 1978](#)), lake delta expansion may still be actively storing sediment to some extent. Bank stabilization also is more limited than in the Mississippi in the upper basin (revetments on one bank alternating downstream and no concrete mats). The absence of stabilization is may allow for significant lateral channel migration that may serve as an additional source of suspended sediment in the lower Atchafalaya Basin (as observed between Melville and Morgan City/Wax Lake; [Fig. 3](#)).

5.2. Importance of bed material load transport

The calculated sediment loads in this study do not directly measure bed material (bedload) load, as isokinetic sampling is typically conducted only to 0.9 of total water depth: suspended load is extrapolated for the 0.9–1.0 water depth interval. As discussed in [Gomez \(1991\)](#), the near-bed division between what is in

suspension versus bed material (traction and saltation) load is unresolved. For purposes of this study, an estimate of the relative importance that fraction of bed material load that is involved in translation of channel floor sand dunes (i.e., bedform transport) can be calculated using results obtained for the New Orleans to HOP reach in Nitttrouer et al. (2008) and Allison and Meselhe (2010). These studies utilized repeat multibeam bathymetric mapping of bank-to-bank dune translations to derive “bedform” transport rates. The resulting exponential relationship ($r^2 = 0.98$) of bedform transport rates with water discharge is:

$$y = 146.3e^{1.77E-4 \times x} \quad (1)$$

where y = bedform transport rate in metric tons/d and x = water discharge (cms). An estimate for the bedform (sand) transport rate for the Belle Chasse station can be calculated from daily water discharge and Eq. (1)—Belle Chasse is the only station that falls within the study reach of the bedform transport studies and limited spatial variability in these rates has been detected in this reach. Annual bedform transport rates were 2.3 MT, 1.5 MT and 2.2 MT in FY08, 09, and 10, respectively. No sand was observed in suspension (using the sediment rating curve) at Belle Chasse at water discharges below about 10,600 cms. These conditions existed, on average, for 82 days/y over FY08–10. Bedform transport continued even at these low discharges (100% of total sand transport; Nitttrouer et al., 2008), but averaged only 0.06 MT/y over this period (3% of the average annual bedform transport total). At a moderate discharge of 535,000 cfs (15,150 cms), sand transport as bedform load was 7.8% (2137 tons/d) of suspended sand transport: that value rose to 12.5% (21,977 tons/d) at a flood of 1,000,000 cfs (28,317 cms). Averaged over FY08–10, sand transport through bedform translation was 2.2 MT/y and suspended sand transport (Table 1) was 17.8 MT/y at Belle Chasse.

We conclude that bedform sand transport was of sufficient magnitude to partly (<14% of the 8.5 MT/y; Fig. 7) account for the interannual channel bed storage measured in the Belle Chasse to HOP reach. This interannual channel storage is assumed to be primarily sand given the results of recent bed surveys (Allison and Meselhe, 2010; Nitttrouer et al., 2011b). While these results cannot be extrapolated upriver to examine its impact on the sand storage in the Tarbert Landing to Baton Rouge reach, the close coupling of bedform transport rates with stream power (e.g., exponential relationship with water discharge; Richards, 1982) suggest this transport mechanism is closely linked with changes in the water surface slope related to backwater flow and to the loss of water at ORCS.

5.3. Impact of hysteresis on sediment load calculations

Sediment rating curves utilize a best fit regression to develop a water discharge: sediment load relationship. As such, variability about this best fit curve is assumed to represent sampling and analytical error. In reality, rivers are known to experience hysteresis “loops” that have been attributed to channel bed scour and injection into the suspended load of stored bed sediment throughout the catchment reach (Walling, 1977; Wood, 1977; Meade et al., 1990). In the lower Mississippi, Mossa (1989, 1996) has described the suspended sediment load maxima as preceding the water discharge maxima. However, most of the station data utilized in that examination was based on low temporal resolution sampling (bi-weekly to monthly boat-based sampling). In the lowermost Mississippi and Atchafalaya, this hysteresis is also magnified by low discharge bed storage of fines in the tidal and estuarine reach that are remobilized on the rising limb (Demas and Curwick, 1987; Galler and Allison, 2008; Allison and Meselhe, 2010). The impact of these processes complicates assigning error estimates to the

water discharge: sediment load relationship, making it difficult to separate sample/analytical error from real variability in suspended load.

To evaluate the impact of basin and local remobilization on the daily suspended sediment loads derived for the monitoring stations in the present study (Fig. 3), we calculated suspended sediment loads for three stations (e.g., Baton Rouge, Belle Chasse and RK39) on the lower Mississippi where optical backscatterance sensors (OBS) were installed to measure turbidity (NTU) for a significant fraction of FY08–10. Methods for calibrating these sensors are reported in the Supplementary file and the results are plotted in Fig. 9. This calibrated load represents a total suspended load and cannot be divided into sand and mud fractions. These records demonstrate that most peaks in water discharge were accompanied by a rising-limb maximum in total sediment discharge of less than 1 week duration, with suspended loads falling prior to peak water discharge. An exception to this trend was observed in late June–early July in both 2009 and 2010 (Fig. 9 arrows); this period coincided in both years with late, snowmelt-related increases in water discharge from the Missouri River, which is known to be the predominant sediment source in the Mississippi River (Meade and Moody, 2010). When compared to rating curve results, the OBS total suspended load at Baton Rouge was 200 MT integrated over 727 days, while the rating curve for those days was 191 MT. At Belle Chasse over 391 days, the OBS was 117 MT and the rating curve 120 MT. These daily results demonstrate that the rating curve method utilized for all the stations in the present study is an accurate method for deriving total suspended discharges when averaged over annual timescales. However, hysteresis was clearly a major factor when examining sediment discharges over shorter time intervals, and the water: sediment discharge relationship is more complicated than reported in previous studies (e.g., Mossa, 1996). By seasonally altering the timing of maximum sediment discharge relative to the water discharge peak, this phenomenon has major implications for the supply of sediment to the Gulf of Mexico (see below) and operation strategies for existing and future river diversions in the lowermost Mississippi to optimize sediment delivery to wetlands.

5.4. Mississippi–Atchafalaya sediment discharge to the Gulf of Mexico

While the Morgan City and Wax Lake monitoring stations are within the tidal reach of the Atchafalaya River and relatively close to the outlet to the Gulf of Mexico (e.g., Atchafalaya Bay), the lowermost station on the Mississippi at Belle Chasse is 121 km upstream of the entrance (HOP; Fig. 1b) to the deep-water passes to the Gulf of Mexico. Our quantification of water and sediment loss through the natural passes and man-made exits, and the channel aggradation in the Belle Chasse to HOP reach, has a major impact on the water and sediment loads exiting to the deeper Gulf. Water loss through the exits above HOP amounted to 44.8% of the annual water discharge reaching Belle Chasse. With the addition of loss to channel aggradation, sediment loss above HOP amounted to 37.9 MT/y, or 43% of the total suspended load passing Belle Chasse. Given that recent surveys have indicated bottom morphology in the Belle Chasse-to-HOP reach is lateral sand bars, relict fluvio-deltaic outcrops, and limited recent fines in the shallows that are seasonally deposited and reworked (Allison and Meselhe, 2010; Nitttrouer et al., 2011b), we conclude that most of the sediment loss to channel aggradation was sand. This loss is as much as 13.1 MT/y of sand loss through a combination of channel storage and exits—equivalent to 74% of the sand load passing Belle Chasse (Fig. 7). This quantity would be reduced if significant mud was disseminated in the sand-rich channel deposits, but recent bed sampling confined to the bar surfaces in the reach below Belle Chasse indicates near surface sands are less than 10% mud at

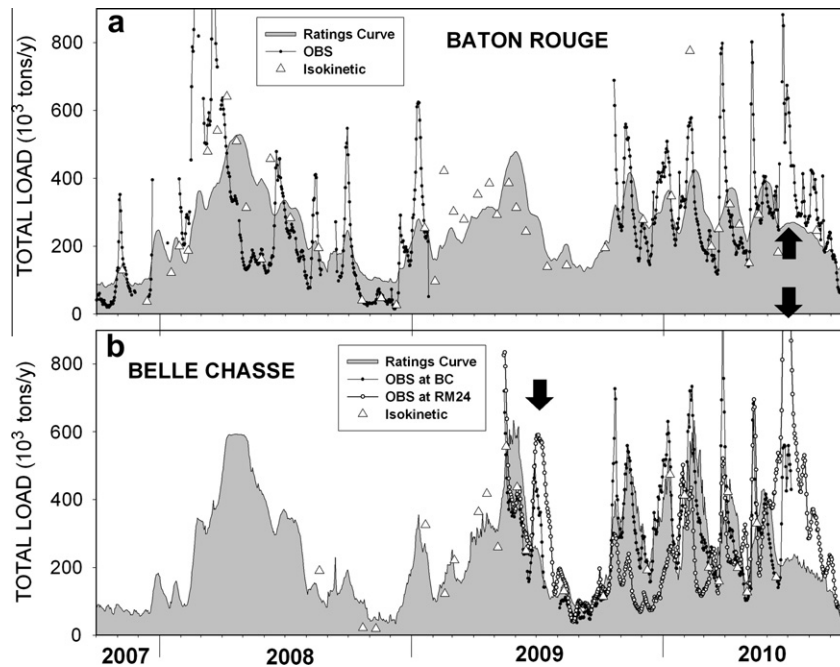


Fig. 9. Total suspended sediment load in the Mississippi River at Baton Rouge (a) and Belle Chasse, Louisiana (b) for flood years 2008–2010 as measured by (1) boat-based isokinetic sampler, (2) optical backscatterance (turbidity) sensor, and (3) calculated ratings curve from the isokinetic measurements and daily water discharges. The optical backscatterance sensor (OBS) records on the Belle Chasse plot include one sensor at the Belle Chasse site and another at river mile 24 (location on Fig. 1b). Arrows indicate late June–early July increases in sediment load interpreted in the text as being due to an increased Missouri River tributary contribution.

high and low discharge (Allison and Meselhe, 2010). A more likely possibility would be if bed material transport plays a significant role in supplying the sand for channel aggradation (Section 5.2).

Several methods were pursued to validate the quantities of water and suspended sediment loss below Belle Chasse. The OBS record at RK39 (Fig. 1b) was converted to an annual total suspended load (see Supplementary file for methods) using Belle Chasse minus Bohemia water discharge (Fig. 6). The results in Table 1 averaged 86.3 MT/y, which agrees well with the 88.0 MT/y for Belle Chasse minus Bohemia in Fig. 6.

Other independent calculations of in channel water and sediment discharge were less in agreement with the budgets shown in Figs. 6 and 7. Boat measurements were made at RK4.2 and RK8.4/15.3 (Fig. 1b; Table 1) by the authors (Pratt) in FY08–10. Details of these methods can be found in the Supplementary file, but the water rating curve developed for RK4.2 averages 277 cubic km/y, compared with 291 cubic km/y for Belle Chasse minus measured exits above HOP (Fig. 6). The 3-year averaged annual water discharge at RK8.4/15.3 was 340 cubic km/y compared with 386 cubic km/y for Belle Chasse minus measured exits above RK8.4/15.3. Water discharge measurements and independent rating curves for the three pass entrances below HOP (used to calculate values in Fig. 6) totaled 253 cubic km/y, while the Belle Chasse minus measured exits above HOP was 291 cubic km/y. These three underestimations of the water discharge are the primary cause for reduced sediment loads calculated by rating curves (Table 1) for these cross-sections. An additional source may be the use of a lighter (P-61) sampler that produced larger wire angles. For instance, the 3-year averaged annual sediment discharge at RK8.4/15.3 was 30.6 MT/y (2.4 MT/y of sand), compared with 51.1 MT/y (12.7 MT/y of sand) for Belle Chasse minus measured exits above RK8.4/15.3.

ADCP measurement error (which typically averages 1–2% of cross-sectional discharge; Edwards and Glysson, 1988) cannot explain this apparent water deficit. While factors such as loss to

groundwater/evaporation, limited number of data points, and underestimation of the water loss through the poorly constrained exits at Bohemia, Ostrica, and Ft. St. Philip may have played a role, we believe tidal and estuarine salt wedge processes were the most significant variable. Diurnal tidal modulation of water velocity produces a variation in ADCP cross-sectional discharge measurements made over different phases of the tide; and this becomes a more important process approaching the Gulf and at lower discharges (Meade, 1972; Galler and Allison, 2008). Even more important to discharge measurements made at low flows and utilized to construct the water and sediment ratings curves for RK4.2 and RK8.4/15.3, a persistent salt wedge is present in the main channel above HOP at discharges below about 8500 cms (Soileau et al., 1989) and is typically confined to the reach below Baptiste Collette except in extreme low water periods. Below the pycnocline, water is moving upstream and the density interface itself serves to degrade ADCP measurement quality of this deep layer. Further, the estuarine circulation processes trap and store fine-grained sediments on the channel floor under the wedge and, through flocculation, reduce turbidities in the saline layer (Galler and Allison, 2008). The latter also will impact sediment load rating curves. Given that the passes in the wedge-influenced reach (e.g., Cubit's Gap, Grand Pass, etc.) have entrance channels perched above the main channel floor, the salt wedge is not present in these channels. We conclude the methods followed to produce the budgets in Figs. 6 and 7 are the most reliable, although they are likely impacted by the tidal modulation, suggesting the need for a more comprehensive monitoring of water and sediment flow through these exits, by future studies, to refine the budget.

While stage measurements to follow water surface slopes described in Section 5.1 are difficult to examine in the reach below Belle Chasse because of a limited number of gauges that record in sufficiently rapid frequency to remove diurnal tidal-induced stage variation, and an overall limited stage range, we can infer that a reduction in stream power, and sediment transport capacity,

takes place associated with the water exits. Stream power is potential energy reduction per unit river length (Lane, 1955; Knighton, 1998; Biedenbarn et al., 2000):

$$\Omega = \gamma QS \quad (2)$$

where Ω = stream power per unit channel length, γ = specific water weight, Q = discharge and S = channel slope. Since specific weight is relatively constant, Ω is proportional to QS . While Nitttrouer et al. (2011a) demonstrated that there is an increase in slope in this backwater reach associated with high water discharges, it can be anticipated that this increase in slope power is reduced by the removal of discharge through the exits above HOP. This mechanism is the most likely explanation for the observed progressive downstream increase in channel storage (Fig. 8). The potential increase in shoaling and its implications for navigation of deep draft vessels must be considered for future sediment (land building) diversions for coastal restoration in the adjacent interdistributary basins (Breton Sound, Barataria Basin; Fig. 1b). A secondary factor could be the ongoing relative sea level rise (RSLR) in south Louisiana, which represents a combination of land subsidence in conjunction with sea level rise (Blum and Roberts, 2009 and references therein). The potential role of this factor in channel storage (by the creation of accommodation space) is not definable at present due to a limited understanding of regional trends in subsidence. However, we believe that this factor is secondary given that present RSLR estimates for the area by Blum and Roberts are ≤ 1.5 cm/y. In comparison, the 1992–2003 volumes of sediment storage in the reach below Belle Chasse, when converted to an average vertical accretion over the study areas (km^2), ranged from 4.3 to 13.5 cm/y for the reaches in Fig. 8. If subsidence impacts the sand storage, it would increase the mass stored in the channel that we have calculated, since the creation and filling of accommodation space would not be marked by a shoaling in elevation.

A summation of the channel/overbank storage of Mississippi + Red suspended sediment in the Atchafalaya and Mississippi channels indicate as much as 44% of the annual total suspended sediment load (100 of 229 MT/y) and 80% (69.2 of 86.0 MT/y) of the sand load was sequestered between the Old River Control Structures and the Mississippi–Atchafalaya exits to the Gulf of Mexico in FY08–10. An additional 34% of the suspended sediment (26% of the sand; indicating the passes are less efficient at discharging sand) exited through man-made and natural passes below Belle Chasse in the Mississippi and did not directly debouch into the deep water Gulf of Mexico (Fig. 7). Together, channel/overbank storage and exit loss mean that only 19% of the annual total suspended sediment load (30.3 of 157.0 MT/y) and 2% (1.6 of 88.3 MT/y) of the sand load at Tarbert Landing reached the deep water Gulf via the main pathway in FY08–10. These figures are equivalent to (or exceed) the fraction of sediment load stored in the LDE reach of other large rivers whose sediment budget has been quantified. In the Amazon, a quantification of coastal and marine depocenters, led Nitttrouer et al. (1995) and Mertes et al. (1996) to conclude that about 1/3rd of the total suspended load of this relatively unaltered river was stored in the LDE reach, primarily by overbank sedimentation. A similar fraction has been attributed to LDE overbank storage in more heavily altered (e.g., artificial levees) Ganges–Brahmaputra delta (Allison et al., 1998; Goodbred and Kuehl, 1999; Kuehl et al., 2005). The lower Mississippi–Atchafalaya storage is unusual for two reasons. The storage and overbank aspects of the sediment budget were quantified from direct measurement of changing water and suspended sediment loads at multiple stations along the river (except for the channel aggradation analysis). The density of monitoring stations in most large rivers is insufficient to allow for this type of analysis. Secondly, this storage is taking place in a system that has been highly altered for flood control, and stands in the future, to be further altered by artificial river diversions that will magnify a process (e.g., crevasse

splay deposition) that, along with overbank flow, plays a primary role in sequestering sediment in LDE's landward of the global ocean.

6. Conclusions

The lowermost Mississippi and Atchafalaya system (e.g., LDE) is generally thought to be an efficient conveyor for throughput of basin sediment to the Gulf of Mexico although the relatively young Atchafalaya distributary has been known to be aggrading historically. Guide levees have cut off overbank flooding to much of the pre-levee lowland floodplain and deltaic plain areas. Bank stabilization practices throughout the channel length (including upriver of the study reach) limit lateral channel migration that remobilizes alluvial deposits. In spite of these control practices, the results of the present study indicate that the LDE zone in the Mississippi–Atchafalaya was as effective at trapping fluvial sediment in FY08–10 as other large major rivers, such as the Amazon and Ganges–Brahmaputra. As much as 44% of the annual total suspended sediment load and 80% of the sand load of the Mississippi + Red Rivers was sequestered between the Old River Control Structures and the Mississippi–Atchafalaya exits to the Gulf of Mexico in FY08–10. This reduction in available suspended sediment is a significant consideration for coastal restoration in the Mississippi delta to use diversions from the channel to create land-building splay deposits in the adjacent interdistributary basins—particularly the removal of sand from suspension that is vital, in our opinion, to establishing marsh platform substrates. Much of the planning to date has relied on suspended sediment availability estimates taken from long-term stations upstream such as Tarbert Landing. The results of the present sediment budget suggest that only a relatively small proportion of the upstream sediment load is available for coastal restoration approaching the Gulf.

Another serious implication of the present study for channel diversions is the linkage between water exits and downstream aggradation of the river channel. There was a progressive increase in channel aggradation rates downstream between 1992 and 2003; sequestering as much as 48% of the suspended sand reaching Belle Chasse, depending on the contribution of bed material load. It is the Belle Chasse to Head of Passes reach that is the focus of diversion planning. The reduction in stream power is attributed to multiple natural pass and man-made exits below Belle Chasse that, in addition to being linked to channel bed aggradation, collectively removed 45% of the water and 34% of the available suspended sediment (26% of the sand) in this reach in FY08–10. These large sediment inputs into the lower parts of the shallow Breton Sound and Barataria interdistributary basins are inefficient at land-building—based upon land-loss studies of these regions conducted by other investigators. These results suggest the need to (1) locate diversions above the rapidly subsiding thick Holocene section near the river mouth and (2) limit diversion water discharge in order to reduce potential channel aggradation that imperils navigation by deep-draft vessels. The latter limitation could be minimized by operating diversions at maximum during the sediment-rich rising limb of the hydrograph and minimizing loss in stream power during the flood maximum which is most efficient at bed material transport and remobilizing suspended sediments deposited during the low discharge phase.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.jhydrol.2012.02.020.

References

- Allison, M.A., Meselhe, E.A., 2010. The use of large water and sediment diversions in the lower Mississippi River (Louisiana) for coastal restoration. *J. Hydrol.* 387, 346–360.
- Allison, M.A., Kuehl, S.A., Martin, T.D., Hassan, A., 1998. The importance of floodplain sedimentation for river sediment budgets and terrigenous input to the oceans: insights from the Brahmaputra-Jamuna River. *Geology* 26, 175–178.
- Barras, J., 2006. Land area Change in Coastal Louisiana after the 2005 Hurricanes: A Series of Three Maps. US Geological Survey Open File Report 06-1274.
- Barras, J., Beville, S., Britsch, D., Hartley, S., Hawes, S., Johnston, J., Kemp, P., Kinler, Q., Martucci, A., Porthouse, J., Reed, D., Roy, K., Sapkota, S., Suhayda, J., 2003. Historical and Projected Coastal Louisiana Land Changes: 1978–2050. US Geological Survey Open File Report 03-334, 39 p. (revised January 2004).
- Baumann, R.H., Day, J.W., Miller, C.A., 1984. Mississippi deltaic wetland survival: sedimentation versus coastal submergence. *Science* 224, 1093–1095.
- Beverage, J.P., 1987. Determining True Depth of Samplers Suspended in Deep, Swift Rivers. US Federal Interagency Sedimentation Project Report GG, 56 pp.
- Bianchi, T.S., Allison, M.A., 2009. Large-river delta-front estuaries as natural “recorders” of global environmental change. *Proc. Natl. Acad. Sci. USA* 106, 8085–8092.
- Biedenharn, D.S., Thorne, C.R., 1994. Magnitude frequency analysis of sediment transport in the Lower Mississippi River. *Regul. Rivers: Res. Manage.* 9, 237–251.
- Biedenharn, D.S., Thorne, C.R., Watson, C.C., 2000. Recent morphological evolution of the Lower Mississippi River, USA. *Regul. Rivers: Res. Manage.* 13, 517–536.
- Blum, M.D., Roberts, H.H., 2009. Drowning of the Mississippi Delta due to insufficient sediment supply and sea level rise. *Nat. Geosci.* doi:10.1038/ngeo553.
- Boesch, D.F., Josselyn, M.N., Mehta, A.J., Morris, J.T., Nuttle, W.K., Simenstad, C.A., Swift, D.J.P., 1994. Scientific assessment of coastal wetland loss, restoration and management in Louisiana. *J. Coastal Res. Special Issue* 20, 103 p.
- Chaudhry, M.H., 2008. Open-Channel Flow. Springer, New York, 523 p.
- Demas, C.R., Curwick, P.B., 1987. Suspended-sediment, bottom-material, and associated chemical data from the lower Mississippi River, Louisiana. Basic records report #14. Baton Rouge, Louisiana, U.S. Geological Survey, Louisiana Department of Transportation and Development, 117 p.
- Edwards, T.K., Glysson, G.D., 1988. Field Methods for Measurement of Fluvial Sediment. US Geological Survey Open-File Report 86-531.
- Galler, J.J., Allison, M.A., 2008. Estuarine controls on fine-grained sediment storage in the lower Mississippi and Atchafalaya Rivers. *Geol. Soc. Am. Bull.* 120, 386–398.
- Galler, J.J., Bianchi, T.S., Allison, M.A., Campanella, R., Wysocki, L., 2003. Sources of aged terrestrial organic carbon to the Gulf of Mexico from relict strata in the Mississippi River. *EOS, Trans. Am. Geophys. Union* 84, 469–476.
- Gomez, B., 1991. Bedload transport. *Earth Sci. Rev.* 31, 89–132.
- Goodbred, S.L., Kuehl, S.A., 1999. Holocene and modern sediment budgets for the Ganges–Brahmaputra River: evidence for highstand dispersal to flood-plain, shelf, and deep-sea depocenters. *Geology* 27, 559–562.
- Harmar, O.P., Clifford, N.J., Thorne, C.R., Biedenharn, D.S., 2005. Morphological changes in the lower Mississippi River: geomorphological response to engineering intervention. *River Res. Appl.* 21, 1107–1131.
- Horowitz, A.J., 2010. A quarter century of declining suspended sediment fluxes in the Mississippi River and the effect of the 1993 flood. *Hydrol. Process.* 24, 13–34.
- Hupp, C.R., Demas, C.R., Kroes, D.E., Day, R.W., Doyle, T.W., 2008. Recent sedimentation patterns within the central Atchafalaya Basin, Louisiana. *Wetlands* 28, 125–140.
- Knighton, D., 1998. Fluvial Forms and Processes. John Wiley & Sons, New York, NY.
- Kuehl, S.A., Allison, M.A., Goodbred, S.L., Kudrass, H., 2005. The Ganges–Brahmaputra delta. In: Giosan, L., Bhattacharya, J. (Eds.), *River Deltas: Concepts, Models and Examples*, vol. 83. Society for Sedimentary Geology (SEPM) Special Publication, pp. 413–434.
- Lane, E.W., 1955. The importance of fluvial morphology in hydraulic engineering. *Proc. Am. Philos. Soc.* 123, 168–202.
- Meade, R.H., 1972. Transport and Deposition of Sediments in Estuaries. In: Nelson, B.W. (Ed.), *Environmental Framework of Coastal-plain Estuaries: Geological Society of America Memoirs*, vol. 133, pp. 91–120.
- Meade R.H., 1995. Setting: Geology, hydrology, sediments, and engineering of the Mississippi River. In: Meade, R.H. (Ed.), *Contaminants in the Mississippi River*, 1987–92. US Geological Survey Circular, vol. 1133, pp. 12–28. <<http://pubs.usgs.gov/circ/circ1133>>.
- Meade, R.H., 1996. River-sediment inputs to major deltas. In: Milliman, J., Haq, B. (Eds.), *Sea-Level Rise and Coastal Subsidence*. Kluwer, London, pp. 63–85.
- Meade, R.H., Moody, J.A., 2010. Causes for the decline in suspended-sediment discharge in the Mississippi River system, 1940–2007. *Hydrol. Process.* 24, 35–49.
- Meade, R.H., Parker, R.S., 1985. Sediment in Rivers of the United States. National Water Summary 1984. US Geological Survey Water-Supply Paper No. 2275, pp. 49–60.
- Meade, R.H., Yuzyk, T.R., Day, T.J., 1990. Movement and storage of sediment in rivers of the United States and Canada. In: Wolman, M.G., Riggs, H.C. (Eds.), *Surface Water Hydrology*, vol. O-1, Geological Society of America, The Geology of North America: Boulder, Colorado, pp. 255–280.
- Mertes, L.A.K., Dunne, T., Martinelli, L.A., 1996. Channel-floodplain geomorphology along the Solimões–Amazon River, Brazil. *Geol. Soc. Am. Bull.* 108, 1089–1107.
- Meybeck, M., Vörösmarty, C.J., 2005. Fluvial filtering of land to ocean fluxes: from natural Holocene variations to Anthropocene. *C. R. Geosci.* 337, 107–123.
- Milliman, J.D., Meade, R.H., 1983. World-wide delivery of river sediment to the oceans. *J. Geol.* 91, 1–21.
- Mossa, J., 1989. Hysteresis and non-linearity of discharge-sediment relationships in the Atchafalaya and lower Mississippi rivers. In: *Sediment and the Environment. Proceedings of the IAHS Symposium*, IAHS Publication, vol. 184, Baltimore, MD, pp. 105–112.
- Mossa, J., 1996. Sediment dynamics in the lowermost Mississippi River. *Eng. Geol.* 45, 457–479.
- National Academy of Sciences (NAS), 2008. First Report from the NRC Committee on the Review of the Louisiana Coastal Protection and Restoration (LACPR) Program. National Research Council of the National Academies. The National Academies Press, Washington, DC, 27 pp.
- Nittroter, C.A.S.A., Kuehl, C.A., Sternberg, R.W., Figueiredo, A.G., Faria, L.E.C., 1995. An introduction to the geological significance of sediment transport and accumulation on the Amazon continental shelf. *Mar. Geol.* 125, 177–192.
- Nittroter, J.A., Allison, M.A., Campanella, R., 2008. Evaluation of bedload transport in the lower Mississippi River: implications for sand transport to the Gulf of Mexico. *J. Geophys. Res. – Earth Surf. Process.* 113, F03004. doi:10.1029/2007JF000795.
- Nittroter, J.A., Mohrig, D., Allison, M.A., 2011a. Punctuated sand transport in the lowermost Mississippi River. *J. Geophys. Res. – Earth Surf. Process.* 116, F04025. doi:10.1029/2011JF002026.
- Nittroter, J.A., Mohrig, D., Allison, M.A., Peyret, A.B., 2011b. The lowermost Mississippi River: a mixed bedrock-alluvial channel. *Sedimentology*. doi:10.1111/j.1365-3091.2011.01245.
- Parker, G., 2004. 1D Sediment Transport Morphodynamics with Applications to Rivers and 905 Turbidity Currents: E-book, St. Anthony Falls Laboratory, University of Minnesota, Minneapolis, p. 906. <http://vtchl.uiuc.edu/people/parker/morphodynamics_e-book.htm> (5.08.10).
- Penland, S., Ramsey, K., 1990. Relative sea-level rise in Louisiana and the Gulf of Mexico: 1908–1988. *J. Coastal Res.* 6 (2), 323–342.
- Richards, K., 1982. Rivers, Form and Process in Alluvial Channels. Methuen & Co., New York, NY.
- Rouse, L.J., Roberts, H.H., Cunningham, R.H., 1978. Satellite observation of subaerial growth of Atchafalaya Delta, Louisiana. *Geology* 6, 405–408.
- Saucier, R.T., 1994. Geomorphology and Quaternary Geologic History of the Lower Mississippi Valley, US Army Corps of Engineers, Mississippi River Commission, Vicksburg, MS, vol. 1, 364 p.
- Simons, D.B., Stevens, M.A., Duke Jr., J.H., 1973. Predicting stages on sand-bed rivers. *J. Waterways, Harbours, Coast. Eng. Div., Am. Soc. Civil Eng.*: New York, 1–9.
- Soileau, C.W., Garrett, B.J., Thibodeaux, B.J., 1989. Drought induced saltwater intrusion on the Mississippi River. In: Magoon, O.T., et al. (Eds.), *Proceedings of the Sixth Symposium in Coastal and Ocean Management*, pp. 2823–2836.
- Swift, D.J.P., 1994. Scientific assessment of coastal wetland loss, restoration and management in Louisiana. *J. Coast. Res. Spec. Issue* 20, 103.
- Syvitski, J.P.M., Vörösmarty, C.V., Kettner, A.J., Green, P., 2005. Impact of humans on the flux of terrestrial sediment to the global coastal ocean. *Science* 308, 376–380.
- Tye, R.S., Coleman, J.M., 1989. Evolution of Atchafalaya lacustrine deltas, south-central Louisiana. *Sed. Geol.* 65, 95–112.
- Vörösmarty, C.V., Fekete, B.M., Meybeck, M., Lammers, R.B., 2000. Geomorphometric attributes of the global system of rivers at 30-minute spatial resolution. *J. Hydrol.* 237 (1–2), 17–39.
- Vörösmarty, C.V. et al., 2004. Humans transforming the global water system. *EOS Trans. Am. Geophys. Union* 85, 509–520.
- Walker, H.J., Coleman, J.M., Roberts, H.H., Tye, R.S., 1987. Wetland loss in Louisiana. *Geogr. Ann.* 69A, 189–200.
- Walling, D.E., 1977. Assessing the accuracy of suspended sediment rating curves for a small basin. *Water Resour. Res.* 12, 1869–1894.
- Walling, D.E., Fang, D., 2003. Recent trends in the suspended sediment loads of the world's rivers. *Global Planet. Change* 39, 111–126.
- Wood, P.A., 1977. Controls of variations in suspended sediment concentration in the River Rother, West Sussex, England. *Sedimentology* 24, 437–445.